PolicyCut — Policy-Driven Local Privacy Video Editor

AMD Radeon Hackathon 2026 · Track 1 · Multimodal Content Creation Tools

1. Project Background

Video is the default medium for journalism, education, research, corporate communication and customer support — and almost every real-world recording contains personal data: faces of bystanders, license plates, ID documents left on a desk, a monitor showing internal dashboards, a spoken phone number or home address. Publishing or sharing such footage without redaction creates legal exposure (GDPR, PIPL, FERPA-adjacent policies) and real harm to the people captured.

Today's options are poor:

- **Manual editing** (masking objects frame by frame in an NLE) is accurate but takes hours per minute of footage.
- **Cloud redaction services** are fast but require uploading exactly the sensitive footage you are trying to protect — often a non-starter for newsrooms, schools, hospitals and research labs.
- **Single-model demos** (a face blurrer, a transcript scrubber) each cover one modality with no review step, no audit trail and no way to express *what* should be redacted.

PolicyCut closes this gap: a fully local, human-in-the-loop privacy editor. The user writes a privacy policy in plain English — *"Pixelate every face, license plate, identity document and screen. Beep phone numbers, email addresses and street addresses."* — and PolicyCut turns it into structured rules, finds the targets in both the video and the audio track, lets a human confirm, correct and complete the machine's proposal, then renders a redacted MP4 with redacted subtitles and a machine-readable audit report. Every model runs locally on an AMD Radeon GPU via ROCm; no frame, no audio sample and no transcript ever leaves the machine.

2. Target Users & Application Scenarios

User	Scenario
Newsrooms & documentary teams	Blur bystander faces and plates in street footage before broadcast; beep spoken contact details of sources
Schools & universities	Redact students' faces and IDs in classroom or campus recordings before sharing with parents or on public channels
Research labs	De-identify study recordings (interviews, driving studies) before archiving or data sharing, with an audit file for the ethics board
Enterprises	Scrub screens, badges and spoken customer PII from meeting recordings, screen captures and support-call videos
Public sector	Prepare bodycam / CCTV excerpts for disclosure with a verifiable record of what was redacted and why

Common requirements across all of these users: the footage must stay on-premises, a human must remain accountable for the final cut, and the organization needs evidence of the redaction process —

exactly the three properties PolicyCut is built around (local-only, human-in-the-loop, audited).

3. System Architecture

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Browser (vanilla HTML/CSS/JS)
 ├─ Projects / New Project (upload + policy)
 ├─ Processing (SSE progress, stage metrics)
 ├─ Review Workspace (video + canvas overlay, dual-track timeline, event editing)
 └─ Export (render, downloads, audit stats)
          │  REST + SSE (/api/v1)
FastAPI backend (single process, single-GPU job queue)
 ├─ policy/   Qwen3-4B → strict JSON PolicySpec → deterministic keyword fallback
 ├─ media/    ffprobe validation, CFR normalization ≤1080p, analysis proxy ≤1280,
 │            thumbnail, 16 kHz ASR WAV, scene-cut detection
 ├─ vision/   Grounding DINO (open-vocabulary keyframe detection)
 │            + YuNet (small-face CPU assist) → NMS → greedy track association
 │            → SAM 2.1 mask propagation (box-interpolation fallback)
 │            → mask post-processing (fill, dilate, feather, safety-box union)
 ├─ audio/    Whisper large-v3-turbo word-level transcription
 │            → spoken-form normalization ("five five five…" → digits)
 │            → PII localisation (regex + heuristics + optional Presidio)
 ├─ review/   overlay preview renderer (boxes + audio banners burned in)
 ├─ media/render  OpenCV frame effects (blur / pixelate / solid, priority solid>pixelate>blur)
 │            + sample-accurate audio mute/beep (1 kHz, −12 dBFS, 10 ms fades)
 │            → libx264/AAC mux, metadata stripped, ffprobe-validated atomic replace
 └─ audit/    audit.json (models, precision, timings, VRAM, events, review history counts —
              zero PII plaintext), redacted.srt, ZIP bundle
Storage: atomic JSON + compressed NPZ masks under data/projects/<id>/ (no database)
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Analysis pipeline order (sequential model loading to bound VRAM): `NORMALIZING → PARSING_POLICY → TRANSCRIBING → DETECTING_PII → DETECTING_VISUAL → TRACKING → BUILDING_PREVIEW → REVIEW_READY`

The human loop is a first-class stage, not an afterthought: every automatic event starts `pending`; export is blocked until each one is confirmed or ignored. Reviewers can retime events, change effects, re-categorize, add fixed manual regions for missed objects and add manual audio intervals. Every mutation is revision-checked (HTTP 409 on conflict), journaled to `review_history.jsonl`, and immediately marks previous exports as stale.

4. Models & Algorithms

Stage	Model / method	Why this choice
Policy → rules	**Qwen/Qwen3-4B-Instruct-2507**, greedy decoding, strict JSON schema, one repair round, then deterministic keyword parser with negation handling	Small enough to co-reside with vision models on one GPU; the deterministic fallback guarantees a valid PolicySpec 100% of the time
Open-vocabulary detection	**IDEA-Research/grounding-dino-tiny** (Hugging Face Transformers port)	Text-prompted detection covers faces, plates, ID documents and screens with one model; the Transformers port avoids the original CUDA-only custom ops, so it runs unmodified on ROCm
Small-face assist	**YuNet** (OpenCV Zoo, CPU)	Catches small/profile faces below DINO's reliable size; explicitly a CPU assist detector
Mask tracking	**facebook/sam2.1-hiera-small** video propagation, `SAM2_BUILD_CUDA=0` (no CUDA extension; connected-component post-processing in OpenCV)	Pixel-accurate masks between keyframes; small variant keeps VRAM headroom; automatic box-interpolation fallback guarantees no unprotected frames
Speech-to-text	**openai/whisper-large-v3-turbo**, word timestamps	Word-level timing is what makes surgical audio redaction possible; turbo variant keeps real-time factor low
PII localisation	Regex + spoken-form normalization + address heuristics, optional **Microsoft Presidio**	Spoken PII ("five five five … at example dot com") defeats naive regex; normalization with char→word mapping recovers exact timestamps

Key algorithmic details:

- **Keyframe + propagation strategy**: Grounding DINO runs every 0.5 s (YuNet every 0.25 s), SAM 2.1 propagates masks between keyframes — an order of magnitude cheaper than per-frame open-vocabulary detection at equal coverage (measured in `docs/benchmark.md`).
- **Recall-first thresholds** (box 0.25 / text 0.20 / NMS IoU 0.5) tuned on three held-out test clips; misses cost privacy, false positives only cost one reviewer click.
- **Mask safety post-processing**: fill holes → drop specks → dilate 5–12 px → feather 2–4 px → union with the padded detection box, so a drifting mask can never expose the target.
- **Audio safety padding**: −150 ms / +250 ms around every PII span, <200 ms gaps merged; if word timestamps are unavailable the whole segment is redacted and flagged for mandatory review — never silently skipped.

5. AMD Radeon GPU / ROCm Adaptation

PolicyCut runs on an **AMD Radeon PRO W7900 (48 GB) with ROCm** (torch 2.10.0+rocm7.1, HIP 7.1.25424); the full analysis pipeline was executed and measured on this GPU with `--require-rocm` active (see docs/benchmark.md).

1. **All four core models run on the Radeon GPU** — Qwen3-4B (policy), Grounding DINO (detection), SAM 2.1 (tracking) and Whisper large-v3-turbo (ASR) — verified by `scripts/preflight.py` (prints `torch.version.hip`, device name, and a GPU matmul smoke test) and recorded per stage in `audit.json`.

2. `--require-rocm` **contest mode**: core model stages raise instead of silently falling back to CPU, so a "Radeon run" is provably a Radeon run. The only intentional CPU component is the YuNet assist detector (documented as such).
3. **ROCm-clean model choices**: the Transformers port of Grounding DINO (no custom CUDA kernels) and SAM 2.1 with `SAM2_BUILD_CUDA=0` (optional CUDA extension skipped; OpenCV replaces the post-processing op). No Flash-Attention, bitsandbytes, or quantization kernels are required.
4. **Precision policy**: FP16 by default on the W7900; any numerically suspect stage is re-run in FP32 for that stage only, and the fallback is recorded in the audit file.
5. **VRAM discipline**: models load sequentially per stage and are explicitly unloaded (`torch.cuda.empty_cache()`) before the next stage; peak VRAM per stage is measured with `torch.cuda.max_memory_allocated()` around synchronized boundaries and shipped in both the UI and `audit.json`.
6. **Honest metrics**: every stage reports wall time, effective FPS, precision and peak VRAM; GPU timing is bracketed by `torch.cuda.synchronize()`. Reproducible via `scripts/benchmark.py` (results in `artifacts/benchmarks/`).

> *Performance tables (FP16 vs FP32, per-frame vs keyframe+SAM strategy, detection-interval sweep, Whisper RTF, end-to-end real-time factor, peak VRAM) live in* `docs/benchmark.md` *and* `artifacts/benchmarks/`*, produced by the reproducible* `scripts/benchmark.py` *harness — only real measured numbers are ever published there, never estimates.*

6. Outputs & Guarantees

Every export produces four artifacts:

- `redacted.mp4` — H.264/AAC, yuv420p, faststart, input metadata stripped, A/V drift < 100 ms, ffprobe-validated before an atomic replace;
- `redacted.srt` — transcript with sensitive spans replaced by `[REDACTED:PHONE|EMAIL|ADDRESS|MANUAL]` (the raw transcript is never exported);
- `audit.json` — input/output SHA-256, policy, model IDs and precision, per-stage timings and peak VRAM, every redaction event with source (auto/manual) and status, counts of human edits — and **zero PII plaintext, no raw transcript, no absolute paths**;
- `policycut_export.zip` — the three above, bundled.

7. Limitations (stated, not hidden)

PolicyCut is a human-in-the-loop redaction tool, **not** a guarantee of legal compliance. It does not do face recognition/whitelisting, age estimation, speaker diarization, generative inpainting or live capture. Detection is recall-tuned but not perfect — which is precisely why the review workspace, the mandatory confirm/ignore step and manual regions exist. Contest build accepts a single ≤60 s, ≤1080p video per project.

PolicyCut Team — AMD Radeon Hackathon 2026, Track 1. Fully local. Human in the loop. Auditable. Built on AMD Radeon + ROCm.